

VEGETABLES PRODUCT GUIDE

Kharif Season — Telangana & Andhra Pradesh

For Digital Marketing Agency Reference

- 1. Flora — Flowering & Fruit Set Booster (Foliar Spray)**
- 2. Builder — Micronutrient Fertiliser for Flowering (Mg, Zn, P, B)**
- 3. Lamigo — Insecticide (Fruit Borer, Shoot Borer, Caterpillars)**
- 4. Probion — Amino Acid Biostimulant (Growth, Stress Recovery)**

Target Districts: Medak · Rangareddy · Anatapur · Chittoor · Mahabubnagar · Kadapa · Jangareddygudem

PRODUCT 1 — FLORA

Category	Flowering & Fruit Set Booster
Application	Foliar spray
Crops	Vegetables — Tomato, Brinjal, Chilli, Okra, Cucumber, Beans and others
Key Role	More flowers, more fruits, less drop, better nutrition to developing fruit

The Problem Flora Solves

In vegetables, the difference between a good harvest and a poor one is often decided at the flowering stage. Tomato plants produce dozens of flowers — but a large number drop off before becoming fruit. The same happens in chilli, brinjal, okra, and cucurbits. This flower and fruit drop is triggered by heat stress, irregular watering, poor nutrition, or simply the plant not having enough energy to hold all its flowers.

Flora is a foliar spray applied at the flowering stage that solves this directly. More flowers stay on the plant, more fruits set, and those fruits develop better.

When to Apply Flora — Vegetable Stage Guide

Early Flowering

Apply when first flowers appear — maximises fruit set

Fruit Development

Repeat during early fruit stage — improves size and weight

Key Benefits of Flora for Vegetables

① Early and Uniform Flowering

- Flora stimulates more flower buds to form and open at the same time — the plant flowers uniformly
- Uniform flowering means fruits develop together — easier to harvest, better marketable batch
- Earlier flowering gives the crop more time to produce multiple rounds of fruit in the same season

② Reduces Flower and Fruit Drop

- The biggest benefit — less flowers falling means more fruits staying on the plant
- In tomato and chilli especially, flower drop during hot or stressful weather is a major cause of yield loss — Flora fights this directly
- Every flower that stays and becomes fruit is additional income for the farmer

③ Better Fruit Set and Pollination

- More flowers are successfully converted into fruits — the fruit set percentage improves
- Fruits that set are more uniform in size and shape — better appearance for market sale

④ Directs Key Nutrients to Flowers and Fruits

- The fertilisers the farmer is already using work harder — Flora helps the plant use them at the right place and time

Vegetable Crops and Their Specific Benefit

- Tomato: Reduces blossom drop in hot and stressful weather, improves fruit set per cluster, better fruit size and uniform ripening
- Chilli: More flowers retained, higher number of fruits per plant, improved colour development
- Brinjal: Better fruit set, reduces deformed or underdeveloped fruits
- Okra: Uniform pod development, more pods per plant, better pod size
- Cucumber/Gourd: Improves female flower retention, better fruit setting, reduces misshapen fruits
- Beans/Pulses: Better pod fill, reduces pod drop at flowering stage

Simple Way to Understand It

Every dropped flower is a fruit the farmer will never sell. In vegetables where income is counted fruit by fruit, Flora directly adds to the harvest. More flowers staying on the plant, setting into fruits, growing bigger and heavier — that is the Flora effect on vegetables.

PRODUCT 2 — BUILDER

Category	Micronutrient Fertiliser — Flowering & Fruit Set Booster
Key Nutrients	Magnesium (Mg), Zinc (Zn), Phosphorus (P), Boron (B)
Application	Foliar spray — pre-flowering and early fruit development
Crops	Vegetables, Fruits, Horticulture — Tomato, Chilli, Brinjal, Okra and others
Competitor	YARA Bud Builder

What Is Builder and Why Do Vegetables Need It?

Most vegetable farmers focus on NPK — nitrogen, phosphorus, and potassium. But there is a group of nutrients that are equally important at the flowering and fruiting stage: magnesium, zinc, phosphorus, and boron. When these are deficient or unavailable, flowers fail to open properly, pollen is weak, fruit set drops, and the fruits that do develop are small and poor quality.

Builder is a foliar fertiliser that delivers exactly these four nutrients directly to the plant — through the leaves — at the most critical stage. It does not replace soil fertilisers. It acts as a targeted top-up at the exact moment the crop needs them most.

When to Apply Builder — Vegetable Stage Guide

Pre-Flowering Stage	Apply when flower buds are forming — loads the plant with nutrients before flowering
Early Flowering	Apply at first flower opening — supports active pollination and fruit set
Fruit Development	Optional repeat — ensures developing fruits get sustained nutrition

What Each Nutrient in Builder Does

① Magnesium (Mg) — The Energy Carrier

- Magnesium is at the centre of chlorophyll — without it, the plant cannot use sunlight to make food
- At flowering stage, magnesium drives the energy conversion process that powers flower opening and pollen production
- Deficiency causes leaves to turn yellow between the veins — and flowers and fruits to be poorly developed
- Builder delivers a high concentration of magnesium directly through the leaf, bypassing soil availability issues

② Zinc (Zn) — The Hormone Trigger

- Zinc regulates the plant's own growth hormones — particularly auxin, which drives bud and shoot development
- Without enough zinc, flower buds do not develop fully, pollen viability is low, and fruit set is poor

- Zinc deficiency is very common in vegetable crops in Telangana and AP soils — it is one of the most widespread hidden deficiencies
- Builder supplies readily absorbable zinc directly to the leaves when the plant needs it most

③ Phosphorus (P) — The Flowering Fuel

- Phosphorus is critical for energy transfer inside the plant during flowering — it powers the biological reactions that form flowers and fruits
- Also essential for pollen tube growth — the process by which pollination is completed and fruit is formed
- Foliar phosphorus from Builder reaches the flowers directly — faster and more effective than waiting for soil-applied phosphorus to move up

④ Boron (B) — The Pollination Enabler

- Boron is the most critical micronutrient for successful pollination — it supports pollen germination and pollen tube development
- Without sufficient boron, pollen either does not germinate or cannot reach the ovary — flowers open and fall off without setting fruit
- Boron also strengthens cell walls in developing fruits — fruits are firmer, hold better shape, and have longer shelf life
- Boron deficiency is the most common cause of hollow fruit, blossom end issues, and misshapen vegetables

Key Benefits of Builder for Vegetables

- Stronger bud formation — more flower buds open successfully
- Better pollination — pollen is viable and reaches the ovary, fruit sets reliably
- Reduced flower and fruit drop — nutrients support the plant to hold what it produces
- Bigger, better-shaped, heavier fruits — nutrients directed to fruits during development
- Improved fruit colour and shelf life — especially in tomato and chilli
- Works on all vegetables and fruits — tomato, chilli, brinjal, okra, cucumber, beans, leafy vegetables
- Foliar application means faster results — nutrients are absorbed within hours, not days

Simple Way to Understand It

Think of Builder as a targeted meal given to the plant right before its most important moment — flowering. A plant that is well-fed with the right nutrients at flowering produces more flowers, holds them, pollinates them successfully, and grows bigger fruits. Builder gives the plant everything it needs for that critical window.

PRODUCT 3 — LAMIGO

Category	Insecticide — Broad Spectrum
Composition	Chlorantraniliprole 9.3% + Lambda-cyhalothrin 4.6% ZC
Formulation	ZC — Capsule Suspension (slow release, longer lasting)
Crops	Vegetables — Tomato, Brinjal, Chilli, Okra, Beans, Cucurbits and others
Controls	Shoot & Fruit Borers, Caterpillars, Pod Borers, Leaf Eating Worms, Semi-loopers, Leaf Miners
Competitor	Syngenta Ampligo

When to Expect These Pests — Vegetable Stage Guide

These pests appear at different stages in vegetables. Timing the spray correctly with Lamigo is key to protecting the crop:

Leaf Eating Caterpillars

Vegetative stage onward — feed on young leaves, defoliate crop

Shoot Borer

Early growth to flowering — bores into growing tip, kills the shoot

Fruit Borer

Flowering to harvest — enters fruit directly, destroys from inside

Pod Borer / Semi-looper

Flowering & pod fill stage — feeds on pods and leaves together

Understanding Each Pest

① Shoot Borer — కాండం తొలుచు పురుగు

Caterpillar that bores into growing tip — Kaandam Tolachu Purugu

The shoot borer caterpillar bores into the tender growing tip of the vegetable plant. The tip wilts, droops, and dies — called Dead Heart. The plant cannot grow upward normally.

- In brinjal and chilli, shoot borer attack during early growth sets back the entire crop — the plant has to branch sideways and takes weeks to recover
- Dead growing tip means delayed flowering and fewer fruiting branches — directly reducing total yield
- The caterpillar moves from shoot to shoot — one or two per plant can set back the entire crop

② Fruit Borer — కాయ తొలుచు పురుగు

Caterpillar that bores inside fruits — Kaya Tolachu Purugu

The fruit borer is the most economically damaging pest in vegetables like tomato, chilli, and brinjal. The moth lays eggs on flowers and young fruits. The tiny caterpillar bores inside.

- From outside the fruit looks healthy — inside the caterpillar is eating the flesh and seeds
- Infested fruits rot from inside, drop early, or open to reveal hollow, damaged, unsellable produce
- In tomato, a single caterpillar can destroy 4–5 fruits before it is detected — it moves from fruit to fruit
- In chilli, fruit borer causes dark entry holes on the fruit surface — the chilli is unsellable in the market
- Brinjal fruit borer damage leads to complete rejection of the produce at market — total income loss from those fruits

③ **Leaf Eating Caterpillars & Semi-loopers — ఆకు తినే పురుగులు**

Caterpillars that eat leaf area — Aaku Tine Purugulu

Semi-loopers and leaf-eating caterpillars appear in large numbers and consume the leaf surface rapidly. Unlike borers, these are visible on the leaves — but by the time they are noticed, significant leaf area is already gone.

- Each caterpillar eats several square centimetres of leaf per day — large infestations can completely defoliate a crop within days
- Without leaves, the plant cannot make food through photosynthesis — fruit development stops and existing fruits fail to fill properly
- Defoliation during fruiting stage reduces fruit weight and quality significantly

④ **Pod Borer — కాయ తొలుచు పురుగు (Beans, Pulses)**

Pod borer in beans and pulses — damages pods before harvest

- Caterpillar bores into developing bean or pulse pods and feeds on the seeds inside
- Damaged pods turn yellow and drop early, or are rejected at harvest due to entry holes
- Attack at pod fill stage means seeds inside the pod do not develop fully — light pods, poor quality

How Lamigo Works — Dual Mode of Action

Lamigo combines two active ingredients that attack the pest from two completely different directions — making it very difficult for pests to survive or develop resistance:

- Chlorantraniliprole — acts on the insect's muscles. It disrupts calcium balance in muscle cells, causing the caterpillar to go into uncontrolled muscle contractions, stop feeding, and die. Effective through both contact and ingestion. Also works on eggs — eggs do not hatch when the plant surface is treated.
- Lambda-cyhalothrin — acts on the insect's nervous system. Fast knockdown — the insect is affected within minutes of contact. Repels adult moths from landing and laying eggs on treated plants.
- ZC formulation (capsule suspension) — the active ingredients are enclosed in tiny capsules that release slowly. This gives 10–14 days of residual protection — fewer sprays needed compared to conventional formulations.
- Protects at egg, larva, and adult stages — complete pest lifecycle management in one product
- Rainfast within 2 hours of application — effective even in rainy vegetable-growing conditions

Simple Way to Understand It

Fruit borers are the biggest market-value destroyer in vegetables — a tomato or chilli with a hole in it cannot be sold. By the time the farmer sees the hole, the fruit is already ruined. Lamigo stops the caterpillar before it enters — eggs do not hatch, young larvae stop feeding, adults do not land. One spray protects for nearly two weeks.

PRODUCT 4 — PROBION

Category	Amino Acid Biostimulant — Plant Growth & Stress Recovery
Type	Natural protein hydrolysate — rich in free amino acids and peptides
Application	Foliar spray — at transplanting, flowering, fruit set, and stress events
Crops	Vegetables, Fruits, Field Crops — Tomato, Chilli, Brinjal, Okra, Cucurbits and others
Competitor	Syngenta ISABION

What Is Probion and What Does It Do?

Probion is an amino acid biostimulant — made from natural protein that has been broken down into small, plant-ready pieces called amino acids and peptides. Think of it as **pre-digested nutrition**. The plant does not have to spend energy making these amino acids itself — they are delivered directly, ready to use immediately. Every plant naturally produces amino acids — they are the building blocks of all proteins, enzymes, and growth processes inside the plant. **When the plant is stressed — by heat, drought, pest attack, waterlogging, or chemical application** — it loses amino acids faster than it can produce them. Probion replenishes this deficit instantly and gets the plant back on track.

When to Use Probion — Vegetable Stage Guide

Transplanting	Apply immediately after transplanting — reduces transplant shock, faster establishment
Vegetative Growth	Apply at active growth — boosts leaf area, faster canopy development

Flowering

Apply at flower initiation — improves pollination, reduces flower drop

Stress Events

Apply after any stress — hail, heat wave, heavy rain, pest damage, chemical burn

Key Benefits of Probion for Vegetables

① Faster Plant Establishment and Growth

- After transplanting, vegetable seedlings go through stress — roots are disturbed, plant needs time to settle
- Probion applied right after transplanting dramatically reduces this transplant shock — plants recover and start growing within days instead of weeks
- During vegetative growth, amino acids act as direct building blocks — the plant builds leaves, stems, and roots faster
- Greener, denser canopy develops faster — the plant is ready for flowering sooner

② Improves Flowering and Fruit Set

- Specific amino acids in Probion directly support flower formation and pollen development
- Pollen viability improves — better pollination, more flowers successfully converted to fruits
- Reduces flower drop caused by amino acid deficiency — the plant has the resources to hold its flowers

③ Stress Recovery

- After a hail storm, the plant's tissue is damaged and growth stalls — Probion provides the amino acids needed to repair and restart growth quickly
- After heat stress — when temperatures spike and the plant loses water rapidly — Probion helps the plant recover osmotic balance and resume normal function
- **After chemical sprays (herbicides, pesticides) that sometimes cause mild leaf burn — Probion helps the plant recover faster from the chemical stress**
- After pest or disease attacks — the plant uses amino acids to rebuild damaged tissue. Probion accelerates this repair.

④ Improves Nutrient Uptake and Efficiency

- Amino acids act as natural chelating agents — they bind to mineral nutrients (iron, zinc, manganese, copper) in the soil and carry them into the plant more efficiently
- The fertilisers already being applied work better — Probion helps the plant absorb and use them
- Even in soils where nutrients are present but 'locked up' (not plant-available), Probion's amino acids help release and deliver them

⑤ Better Photosynthesis and Overall Plant Health

- Amino acids are directly involved in building chlorophyll — the green pigment that captures sunlight
- Crops treated with Probion show visibly darker green leaves — a sign of better chlorophyll content and more active photosynthesis
- More photosynthesis means more energy available for the plant — better fruit fill, higher sugar content in fruits, improved quality

Where Probion Fits in a Vegetable Spray Program

Probion is not a fertiliser and not a pesticide — it is a biostimulant. It works best when combined with a regular nutrition program. Think of it as a performance enhancer that makes everything else the farmer is doing work better:

- Use at transplanting — reduces crop loss from transplant shock, saves re-planting cost
- Use at flowering — improves fruit set, reduces flower drop
- Use after any stress event — gets the crop back on track faster, reduces yield loss from downtime
- Can be mixed with most fertilisers and pesticides — flexible to add to existing spray schedules

Simple Way to Understand It

When a person is sick or exhausted, a doctor gives them an amino acid supplement — the body absorbs it directly without having to digest anything. Probion does the same for the vegetable plant. When it is stressed, recovering from damage, or about to flower, Probion gives it ready-to-use building blocks to bounce back faster, grow better, and produce more.

PRODUCT 5 — K-MATE

Category	Organic Bio-Stimulant + Soil Conditioner
Key Ingredient	Potassium Humate — high-grade organic potassium from humic sources
Form	Dry black organic granules / powder — applied through drip irrigation
Application	Drip irrigation (primary) — soil application
Crops	Vegetables, Chilli, Tomato and other commercial crops
Like	Similar to Karbac — but specifically designed for drip irrigation use

What Is K-Mate and Why Is It Different from Regular Fertilisers?

K-Mate is built on one powerful ingredient: Potassium Humate. This is potassium that has been organically bound to humic acid — making it completely different from the regular muriate of potash (MOP) or sulphate of potash (SOP) that most farmers use.

Regular potassium fertilisers dissolve in water and move through the soil — but a large portion is either leached away (washed down by irrigation), fixed in the soil (locked up in unavailable forms), or simply not reaching the root zone at the right time. K-Mate solves this by delivering potassium in an organic, soil-active form that stays available longer, improves the soil at the same time, and feeds the plant more efficiently.

It is designed specifically for drip irrigation — so instead of broadcasting it across the field, it goes directly into the root zone through the drip lines. Targeted, efficient, and zero wastage.

When to Apply K-Mate — Vegetable Stage Guide

Transplanting / Land Prep

Apply at the start — conditions the root zone before planting

Vegetative Growth

Continue through growth — builds strong root system and soil health

Flowering & Fruit Set

Continue or increase dose — potassium is critical at this stage

Fruit Development

Keep applying — potassium drives fruit fill, weight, and quality

What K-Mate Does — Explained Simply

① Root Development — Grows Deeper, Stronger Roots

K-Mate's potassium humate stimulates the plant to send roots deeper and wider into the soil. In drip-irrigated vegetables — where the wetting zone is concentrated — stronger vertical root growth means the plant anchors better and taps into deeper moisture reserves during dry spells between irrigations.

- Roots penetrate deeper soil layers — access more water and nutrients than shallow-rooted plants
- More lateral root growth — greater total root surface area absorbs more per irrigation cycle
- Stronger root anchoring — vegetable plants stand upright, resist wind and rain lodging
- Better root-soil contact means the plant responds faster to every fertigation applied through drip

② Soil Conditioning — Makes the Soil Work Better Over Time

One major problem with drip-irrigated vegetable fields is that the soil around the drip zone can become hard and compacted over time — especially with repeated chemical fertiliser use. K-Mate reverses this.

- Humate molecules loosen compacted soil structure — water spreads more evenly through the drip zone
- Improves soil aeration — oxygen reaches the root zone, which beneficial soil microbes need to survive and work
- Increases the soil's capacity to hold nutrients — reduces the leaching of expensive fertilisers below the root zone
- Stimulates beneficial soil microbial activity — a living soil breaks down organic matter and makes nutrients available naturally
- With regular use across seasons, soil structure genuinely improves — less hardpan, better water distribution, lower fertiliser requirement over time

③ Nutrient Uptake — Makes Every Fertiliser Work Harder

K-Mate does not just add potassium — it improves how the plant absorbs all nutrients, including the nitrogen, phosphorus, and micronutrients the farmer is already applying through the drip.

- Humate acts as a natural chelating agent — it binds to nutrients in the soil and carries them into the root in a plant-available form
- Nitrogen uptake improves — the plant uses urea and other nitrogen sources more efficiently, reducing nitrogen loss
- Potassium from K-Mate is in organic humate form — stays in the root zone longer and does not wash away with irrigation water
- Micronutrients like iron, zinc, and manganese that are often locked up in alkaline soils become more available when humate is present

④ Plant Metabolism and Crop Quality

- Potassium is directly responsible for sugar production, fruit sizing, and colour development in vegetables
- K-Mate's potassium drives photosynthesis efficiency — the plant produces more food from the same sunlight
- Promotes chlorophyll formation — crops treated with K-Mate show visibly greener, healthier leaves
- Helps form essential amino acids inside the plant — plant proteins and enzymes are built more efficiently
- Fruit and vegetables grown with regular K-Mate application show better uniform sizing, firmer texture, and improved shelf life — all of which matter for market price

Why Drip Irrigation Is the Best Way to Apply K-Mate

K-Mate is specifically formulated for drip application. Here is why this matters:

- Goes directly into the active root zone — no spreading across the whole field, no wastage
- Dissolves cleanly in drip irrigation water — no clogging of drip emitters
- Applied in small doses with each irrigation cycle — the plant gets a consistent, steady supply rather than a one-time large dose
- Works with any fertigation schedule — can be applied alongside water-soluble NPK fertilisers
- In hot weather between irrigations, potassium from K-Mate helps the plant manage water stress better — stomata close and open more efficiently, reducing water loss

Simple Way to Understand It

Think of K-Mate as a soil doctor that works through the drip pipe. Every time the farmer irrigates, K-Mate goes in — conditioning the root zone, feeding the plant the right potassium, and making the soil better with each cycle. It does not replace the regular NPK program — it makes that program work harder and last longer.

PRODUCT 6 — EXPEL-R

Category	Bio-Stimulant with Pest Control Action
Type	Biological — natural active ingredients, safe for humans and beneficial insects
Crops	Vegetables — Tomato, Brinjal, Chilli, Okra, Beans, Cucurbits

Controls	Lepidopteran caterpillars — Fruit Borer, Shoot Borer, Leaf Eating Worms
Competitor	Shanmukha Agritech Terminator

Who Are These Pests — And Why Do They Matter So Much in Vegetables?

Lepidopteran pests are moths and their caterpillars. In vegetables, it is the caterpillar stage that causes all the damage — and specifically in vegetables, the damage hits the most valuable part: the fruit itself.

What makes these pests especially dangerous in vegetables is timing — they attack exactly when the crop is most valuable: at flowering and fruiting stage. And the damage they cause — holes in fruits — makes the produce completely unsellable even if the rest of the crop is healthy.

① **Fruit Borer — కాయ తొలుచు పురుగు (Tomato, Chilli, Brinjal, Okra)**

Caterpillar that bores inside the fruit — Kaya Tolachu Purugu

When to Expect

From first flower stage onward — peaks at active fruiting stage

The fruit borer moth lays tiny eggs on flowers and young fruits. The newly hatched caterpillar immediately bores into the fruit and starts eating from inside. The outside looks completely normal.

- The farmer cannot see the damage until the fruit is cut or the caterpillar exits — by then the fruit is ruined and unsellable
- In tomato: Entry hole at the top of the fruit, caterpillar eats seeds and inner flesh — fruit rots from inside, falls early or is rejected at market
- In chilli: Round entry hole on the body of the chilli — affected fruits turn red prematurely and fall. Cannot be sold as fresh chilli.
- In brinjal: Caterpillar bores through the stalk into the fruit — entire fruit becomes hollow and black inside
- In okra: Entry near the tip, caterpillar feeds inside the pod — pod becomes deformed and unsellable
- A single caterpillar moves from fruit to fruit — one moth's egg-laying can destroy 8–10 fruits before being detected

② **Shoot Borer — కాండం తొలుచు పురుగు (Brinjal, Chilli)**

Bores into the growing tip — Chivuru Tolachu Purugu

When to Expect

Early vegetative stage — 20 to 60 days after transplanting

In brinjal and chilli, the same moth that later attacks fruits first attacks the growing tips of the plant when the crop is young.

- The caterpillar bores into the tender growing tip — the tip wilts and hangs down (Dead Heart)
- The plant cannot grow upward and branches sideways — delayed development, delayed flowering
- Early shoot borer attack in brinjal can delay the first fruit harvest by 3–4 weeks — significant income loss
- The caterpillar moves from shoot to shoot — in heavy attack, most growing tips in the field are affected

③ **Leaf Eating Caterpillars & Semi-loopers — ఆకు తినే పురుగులు**

Caterpillars that consume leaf area rapidly

When to Expect

Vegetative and flowering stage — any time in the crop cycle

- Feed on leaves rapidly — large infestations defoliate entire plants within days
- Without leaves, the plant cannot make food — fruits on the plant fail to fill and develop properly
- Tomato plants attacked at fruit fill stage produce smaller, lighter fruits — marketable weight drops
- Semi-loopers fold the leaf or hang from it while feeding — visible to the naked eye but population can explode fast

How Expel-R Controls These Pests in Vegetables

Expel-R is a biological product — it uses natural active ingredients that are specifically designed to affect caterpillars (lepidopteran larvae) while leaving the rest of the field ecosystem unharmed.

- When the caterpillar eats plant tissue treated with Expel-R, its gut is disrupted — it stops feeding within hours and dies within 24–72 hours
- The key advantage in vegetables: it stops feeding immediately — even a caterpillar that has just started boring into a fruit stops before it causes further damage
- Repellent effect on adult moths — moths avoid laying eggs on treated plants. This breaks the pest cycle before caterpillars even hatch — the most effective form of control.
- Works on young caterpillars most effectively — early application before the pest bores deep into the fruit is critical. Once the caterpillar is inside the fruit, it is protected from most sprays.
- Safe for pollinators — bees that are essential for fruit set in tomato, chilli, and cucurbits are not harmed
- No harmful chemical residue on vegetables — safe for fresh market sale and export requirements
- Biological mode of action — pests do not develop resistance to Expel-R the way they do to chemical insecticides used repeatedly

Why Expel-R Is Especially Important for Vegetables vs Field Crops

In paddy or cotton, some level of pest damage is accepted and the final product is still marketable. In vegetables, a single caterpillar hole on a tomato or chilli means the entire fruit is rejected at the market. The threshold for damage is zero — one visible hole = zero price.

This is why preventive and early application of Expel-R pays off many times over in vegetables. The cost of one spray is far less than the income lost from even a small percentage of rejected fruits.

Simple Way to Understand It

A fruit borer attack on a chilli or tomato field is not just crop damage — it is market rejection. The farmer grows the fruit, spends money on water and fertiliser, and then cannot sell it because it has a hole. Expel-R stops the moth from laying eggs and stops the caterpillar from entering the fruit — protecting both the crop and the income.